Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery

R26-DS-010

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Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery

1 Problem & Impact

- **Chronic stress, burnout, high error rates** — directly affect patient safety and staff turnover *(Shanafelt et al., Mayo Clinic Proc.)*
- **Compounding burden:** patient falls, manual documentation overload, and communication breakdowns with deaf / hard-of-hearing patients.
- **Existing tools are reactive,** single-modal, and impersonal — missing early-warning signals before burnout threshold is reached.

2 Who Benefits

Tier	Group	Outcome
PRIMARY	Caregivers — nurses, eldercare & home-care staff, and supervisors	Reduced burnout; better performance
SECONDARY	Patients & healthcare institutions	Improved safety, fewer falls, lower turnover
TERTIARY	Deaf and hard-of-hearing patients	Reliable signed communication

3 Design Excellence of the Solution

Unified Multimodal Platform

Physiological + acoustic stress detection, edge-based fall prediction, voice-driven ADL documentation, and adaptive sign-language tutoring.

Personalization Per Caregiver

Isolation Forest baselines and RL-based adaptive learning tuned to each individual caregiver's patterns.

Explainability Built In

Transparent reasoning so caregivers and supervisors can trust and act on the system's signals.

Early-Warning Focus

Proactive — flags risk signals before burnout threshold is reached, unlike existing reactive tools.

Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery

Validation: Field Insights

Remark Elder Care

Discussed "Sign Vital" systems with co-founder **Vidyantha Vasantha**. Identified the need for stress detection via vocal harmonics.

CCHS Colombo

Director highlighted **literacy barriers**. Staff struggle with manual logbooks, requiring hands-free documentation solutions.

Love & Elder Care

Director **Hiran** emphasized that logbook automation and predictive fall analysis are the highest priorities for safety.

IT22178640 | DAYANA PRIYADHARSHANI KUMAR



Component 01: AI-Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution

Specialization : Data Science

AI-Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution

HOW THE SOLUTION ADDRESSES THE SUB-PROBLEM

DETECT	PERSONALISE	PREDICT
Multimodal Stress Detection	**Individual Baseline**	**Burnout Trajectory**
XGBoost on EDA + Temperature + HRV 13,287 windows · 15 real nurses Binary F1 = 0.861 under LOSO CV	Isolation Forest per nurse 7-day calibration · 15 models Ablation: lowest false positive rate	Daily risk score + slope per nurse Days-to-threshold computed Nurse 5C: +2.73/day, already critical

- **3-Class XGBoost (LOSO)** · Macro F1 = 0.305 · Accuracy = 64.1% · 15-fold · No Stress / Medium / High
- **Binary XGBoost (LOSO)** · Binary F1 = 0.816 · Accuracy = 74.7% · RF best at 0.822 → XGBoost selected for exact SHAP
- **Audio Classical (RAVDESS)** · LR best · F1 = 0.780 · AUC = 0.826 · 34 acoustic features · 5-fold actor-independent
- **Audio 1D CNN (RAVDESS)** · F1 = 0.804 · AUC = 0.889 · 130×13 MFCC matrix · selected — best all metrics
- **Late Fusion** · Physio 51.7% + Audio 48.3% · weights derived from validated F1 scores

USER REQUIREMENTS ADDRESSED BY THE SOLUTION

Requirement	Type	Status
Detect stress from physiological signals	*Functional*	DONE
Detect stress from acoustic / voice features	*Functional*	DONE
Personalise alerts to individual baseline	*Functional*	DONE
Show trajectory + forecast per nurse	*Functional*	DONE
Explain model decisions (SHAP per nurse)	*Functional*	DONE
Relational team network graph	*Functional*	PARTIAL
Workload redistribution suggestions	*Functional*	PROTO
Real-time inference backend (FastAPI)	*Functional*	PP2
Accuracy above majority-class baseline	*Non-Functional*	DONE
Per-individual SHAP explainability	*Non-Functional*	DONE
Reduced false positive rate (Isolation Forest)	*Non-Functional*	DONE
Privacy — de-identified · PhysioNet license	*Non-Functional*	DONE

AI-Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution

DESIGN EXCELLENCE / CONTRIBUTION

01 Per-Nurse SHAP Attribution — Addresses Topol (2019) The Lancet gap

GAP: Hosseini (2022) reports only a single global feature importance. Topol (2019) states: AI must explain at the individual level.

CONTRIBUTION: SHAP TreeExplainer run per nurse on LOSO held-out windows → 15 individual profiles. Finding: Nurses 5C/6B/6D = temp_mean dominant (thermoregulatory). Nurses F5/7A/94 = eda_peaks_count dominant (electrodermal). Same model, different explanation per person.

02 Isolation Forest Ablation — Addresses Gjoreski (2017) + Hosseini (2022) gap

GAP: Both papers state personalised thresholds remain an open challenge. No prior paper on this dataset implemented or empirically evaluated it.

CONTRIBUTION: Ablation study: Fixed threshold vs Population baseline vs Isolation Forest. IF achieves lowest false positive rate → fewer false alarms → supervisors trust and act on alerts → reduced alert fatigue.

03 1D CNN on MFCC Temporal Matrix — Novel application context

GAP: Bhangale (2023) applied 1D CNN to 8-class RAVDESS emotion recognition. Not applied to binary occupational stress in a multimodal caregiver system.

CONTRIBUTION: CNN sees full 130-frame × 13 MFCC matrix (not 34 summary stats). Captures temporal vocal stress patterns. Result: F1 = 0.804, AUC = 0.889 — best audio model by all metrics (+0.063 AUC over Logistic Regression).

USER FEEDBACK ON PROTOTYPE

"I can't tell which nurse needs attention at a glance"

✓ Red left-border + Critical badge on high-alert cards. Colour coding: red = HIGH, orange = MODERATE, grey = NONE.

"The risk number alone means nothing to me"

✓ Added Δ vs. personal baseline card (+45.4 for Nurse 5C). Trajectory label (Rising ↑ / Stable / Recovering) + days to threshold.

"What should I actually do when I see a red card?"

✓ Added Workload Redistribution page with approve / modify / reject suggestions — closes the detect → alert → intervene loop.

"Why is this nurse flagged — what caused this?"

✓ Added SHAP chart on every nurse profile. Supervisor sees the exact feature driving risk for that individual.

AI-Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution

Panel Comment Received at Proposal

> "Identify how do you going to identify individual behavioural patterns"

How CareSense Addresses This

- **Layer 1 – Personal Baseline Modeling:** Isolation Forest is trained on each nurse's initial 7-day physiological data to establish an individualized "normal" behavioral and stress baseline.
- **Early Detection of Personal Deviations:** Instead of comparing against population averages, the system identifies abnormal patterns unique to each individual nurse.
- **Layer 2 – Explainable Risk Attribution:** XGBoost combined with SHAP analysis highlights the specific physiological features contributing most to stress risk for each nurse.
- **Personalized Stress Signatures:** Different nurses exhibit different dominant stress indicators (e.g., EDA peaks, temperature variation), enabling tailored interpretation of risk.

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Component 02: Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Observable Fall Risk Detection and Prediction

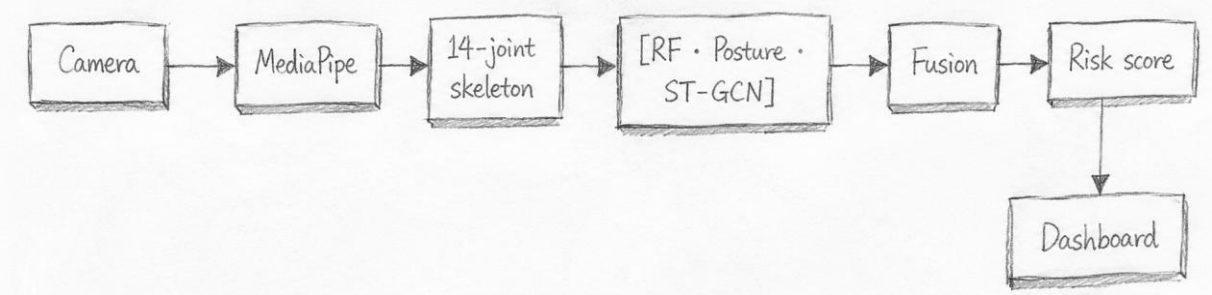
Specialization : Data Science

Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Observable Fall Risk Detection and Prediction

SUB-PROBLEM Existing fall detectors are **binary** (fall / didn't fall) and trigger *after the incident* — missing the build-up to instability and producing high false-alarm rates.

1 My Solution

- **Continuous risk scoring** — Low / Moderate / High instead of binary detection
- **Pipeline:** RGB camera → MediaPipe pose → **14-joint skeleton** → 4 trained models → real-time verdict on edge device (no cloud)
- **Four-model ensemble:**
  - Random Forest baseline (18 hand-crafted motion features)
  - 4-class Posture classifier (Lying / Sitting / Standing / Walking)
  - Spatio-Temporal Graph CNN (ST-GCN, 14-joint motion)
  - Late-Fusion MLP combining all three
- **RGB discarded** at pose extraction — only skeleton processed (privacy-first)
- **Backend ↔ Frontend:** Python pipeline writes JSON manifest → React dashboard reads real model output for every patient card



2 User Requirements Addressed

Requirement	How I address it
Privacy preservation	Skeleton-only data flow; **RGB discarded immediately**
Real-time on consumer laptop	All processing **offline on edge device** — no internet needed
Early warning, not post-fall	Moderate Risk class **fires before fall escalates**
Verifiable, defensible accuracy	**94.24% Fusion test accuracy** on 799 held-out subject-disjoint sequences
Explainable predictions	**Per-patient "Why?" panel** showing dataset action + 8 motion features
Evaluation	**3 protocols** — Cross-Subject, Cross-View, Cross-Dataset — to expose generalisation gaps

Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Observable Fall Risk Detection and Prediction

KEY CONTRIBUTION Late-fusion model combining 4 architectures — **94.24% test accuracy** on 799 held-out subject-disjoint sequences (+6.8% over RF baseline)

1 Design Excellence / Contribution

- **4 models combined:** RF · Posture · ST-GCN · Late-Fusion MLP
- **Edge-first architecture** — inference on Jetson / Coral; only anonymous skeletal vectors leave the camera node
- **Privacy by design** — system never stores or transmits identifiable video
- **Real-time performance** — sub-second end-to-end latency on ward devices
- **Multi-patient tracking** — pose-based identity association handles overlapping subjects in one frame
- **Modular deployment** — swap models (MediaPipe ↔ MoveNet, LSTM ↔ Transformer) without rewriting the system
- **Explainable & integrated** — every prediction shows motion features; backend output drives every dashboard card

2 User Feedback on Prototype

3 stakeholders engaged · **4 changes applied**

Who	What they said	What I changed
Ward Staff	*"No faces leave the room"*	Privacy stance **validated** — no design change needed
Reviewers	*"Show which feature drove a high score"*	Added **explainability layer**
Caregivers	*"Don't alarm us for every brief wobble only sustained instability"*	Added **sustained-warning logic** Moderate Risk fires only after risk stays in band for **12 seconds** (5 ticks); High Risk fires immediately
Advisor	*"Widen the model — sit-to-stand & bed-edge"*	Added **pre-fall behaviors** to feature set

AI-Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution

Panel Comment Received at Proposal

“When it comes to reasoning, skeletal movement might not be enough”

How CareSense Addresses This

- **The system combines deep learning with motion-based reasoning:** A spatio-temporal ST-GCN model captures body movement patterns, while 18 handcrafted motion features provide additional information about human movement and instability.
- **Six reasoning layers improve activity and risk understanding:** The framework analyzes movement over time, detects subtle instability, understands posture states such as sitting or standing, and identifies risky transitions like sit-to-stand or sudden drops
- **Context and pose-quality checks increase reliability:** Spatial context helps interpret movements differently based on location, while the pose-quality gate checks for occlusion or unreliable skeleton detection to reduce false alarms
- **All information is fused into a single final risk score**: The deep learning pose stream and motion features are combined using a Late-Fusion MLP to generate a clear low, moderate, or high risk prediction with better reasoning capability.

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Component 03: Context-Aware Intelligent System for Automated ADL Extraction and Real-Time Documentation from Caregiver Voice Logs

Specialization : Data Science

Context-Aware Intelligent System for Automated ADL Extraction and Real-Time Documentation from Caregiver Voice Logs

1 How the Solution Addresses the Sub-Problem

Sub-Problem	Stage	Solution	Result
Background noise degrades ASR accuracy in care home recordings	Stage 1	noisereduce spectral subtraction 200MB RAM, no GPU required	54.6% RMS reduction
ASR not optimised for Sri Lankan caregiver speech	Stage 2	Whisper-small compared across 3 sizes on 23 test recordings	WER 27.22%
Fillers and self-corrections corrupt downstream NLP processing	Stage 3	BART fine-tuned	BLEU 0.8525
Informal spoken grammar not corrected in ASR output	Stage 4	Same BART v2 handles grammar jointly	ROUGE-L 0.9400
No structured clinical data extracted from voice logs automatically	Stage 5	BERT-NER rule-based extraction domain-specific keyword patterns	F1 91.27%
Patient safety alerts not generated automatically	Stage 6	ClinicalBERT planned	-
Incoming caregivers rely on manual written handover reports	Stage 7	gTTS spoken handover summary from ADL JSON - planned PP2.	Structured report hint in full dataset

2 User Requirements Addressed by the Solution

Requirement	How Addressed
Caregiver must speak naturally without learning new words	BART handles fillers, self-corrections, repetitions in natural Sri Lankan English - no constraints on speaking style
System must work in noisy care home rooms	noisereduce removes fan, AC, and TV noise — 54.6% RMS reduction tested on real care home recordings
Patient hygiene must be captured automatically	BERT-NER extracts hygiene field -partial body wash, full body bath, sponge bath - Precision 100%
Medication timing and dose must be recorded accurately	BERT-NER extracts after_breakfast, after_lunch, after_dinner status and tablet dose for all
Fluid intake must be tracked especially for kidney patients	Fluid extraction accuracy 100% on test rows
Supervisor alerted immediately when medication refused	Alert triggered when medicine refused is detected - Alert Recall 86.05% on test rows
Supervisor alerted when abnormal symptoms are observed	Body pain, fever, vomiting, loose motion, crying detected automatically from cleaned text
System must run on standard care home hardware without GPU	noisereduce 200MB no GPU + BERT-NER ~0MB no GPU - full pipeline CPU-ready for care home deployment

Context-Aware Intelligent System for Automated ADL Extraction and Real-Time Documentation from Caregiver Voice Logs

3 Design Excellence and Contributions

1.First Sri Lankan elder care speech dataset :from 39 patients at Love & Elder Care

2. Deployment-aware model selection :Every model chosen for care home hardware - noisereduce (200MB, no GPU) selected over Facebook Denoiser (4GB, GPU required).

3. Self-correction and repetition resolution and grammar correction Resolves "two tablets no sorry three tablets" → "three tablets" automatically BART

4. Rule-based beats neural -novel finding BERT-NER F1 91.27% vs FLAN-T5 F1 47.57% - domain-specific rules outperform general neural models by 43.7%.

5. Kidney patient special handling fluid_limit_ml tracking and dialysis_status extraction .

6. Automatic SNR measurement Replaced manual noise labels with objective SNR measured from audio - reproducible evaluation independent of human annotation.

4 User Feedback on the Prototype

Who	Feedback	Action Taken
Caregivers	Must speak naturally, no special words	BART v2 handles fillers and self-corrections in natural speech
Supervisor	Medication refusal must alert immediately	Alert Recall 86% — medication refusal is the primary trigger
Supervisor	Kidney patients have fluid limits	fluid_total_ml tracked for kidney patients
Supervisor	Output must be simple, not technical	Human-readable report planned for PP2
Both	Will caregiver listen to a summary?	TTS spoken handover planned for PP2

AI-Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution

Panel Comment Received at Proposal

“Highlight how the proposed methodology differs from a normal ASR model”

How CareSense Addresses This

- **Normal ASR gives only raw text, but my system gives structured ADL records:** A normal ASR model only converts speech into a transcript, but my system extracts ADL details such as medication, food intake, fluid intake, hygiene, symptoms, mobility, and time.
- **Normal ASR keeps speech mistakes, but my system cleans them:** It removes fillers, repetitions, hesitations, and self-corrections
- **Normal ASR does not understand care context, but my system detects risks:** My system identifies missed medication, medication refusal, abnormal symptoms, and kidney patient fluid limit issues.
- **Normal ASR stops at transcription, but my system produces care outputs:** My system generates a structured ADL report, sends supervisor alerts, and creates a handover summary for the next caregiver.

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Component 04: Personalized Sign Language Tutoring System for Caregivers

Specialization : Data Science

Personalized Sign Language Tutoring System for Caregivers

How the Solution Addresses the Sub-Problem

- Provides a **caregiver-focused** sign language tutoring and recognition system for emergency and daily care communication.
- Addresses the lack of **continuous** sign language **practice support** available for caregivers after short-term training programs.
- Uses a custom-selected set of **66 caregiver and emergency-related ASL words** instead of general sign language vocabulary.
- Captures caregiver hand and body movements using **MediaPipe landmark extraction** for real-time gesture analysis.
- Converts raw sign videos into **normalized landmark sequences** for robust and privacy-preserving AI processing.
- Implements deep learning models such as **BiLSTM, GRU, and CNN + BiLSTM** for isolated sign recognition and gesture evaluation.
- Uses **DTW-based motion comparison** to evaluate gesture correctness independently from signing speed.
- Generates **interpretable corrective feedback** to help caregivers identify and improve incorrect sign movements.
- Supports future **adaptive learning** by tracking repeated learner mistakes and personalized practice needs.
- Improves **accessibility and communication support** for caregivers interacting with deaf or hearing-impaired individuals.

User Requirements Addressed by the Solution

- **Easy-to-use** sign language learning support for caregivers
- **Real-time** gesture capture and recognition using standard webcams
- **Accurate recognition** of caregiver-related emergency ASL signs
- **Speed-tolerant** gesture evaluation for different signing speeds
- **Visual corrective feedback** for self-learning and improvement
- **Reliable** landmark-based gesture processing and evaluation
- **Reduced dependency** on interpreters during basic communication
- **Privacy-preserving** processing using skeletal landmark extraction
- **Scalable architecture** for future vocabulary and feature expansion
- **Cross-device accessibility** through browser-based interaction

Personalized Sign Language Tutoring System for Caregivers

Design Excellence and Contributions

1. Caregiver-Focused ASL Tutoring
Focused on caregiver and emergency communication needs validated through CCHS instead of general sign recognition.

2. Custom Emergency Vocabulary
Built a custom 66-word caregiver-focused ASL subset using WLASL and ASL-Citizen datasets.

3. Landmark-Based Processing
Used MediaPipe pose and hand landmarks for lightweight and privacy-preserving gesture processing.

4. Robust Preprocessing Pipeline
Applied normalization, temporal standardization, and corrupted-video filtering for improved model learning.

5. Speed-Tolerant Evaluation
Implemented DTW-based motion comparison to evaluate gestures independently from signing speed.

6. Multi-Model Experimentation
Trained and compared BiLSTM, GRU, and CNN + BiLSTM architectures for caregiver-focused ASL recognition.

7. Corrective Feedback Support
Supports gesture correction and self-learning through motion-aware evaluation and interpretable feedback.

User Feedback on Prototype

Who	What they said	What I changed
Caregivers	*"We only remember a few signs after training."*	Focused the system on a **66-word caregiver and emergency ASL vocabulary** subset for practical communication.
Reviewers	*"Recognition alone is not enough for learning."*	Added **DTW-based gesture evaluation** and **corrective feedback** support.
ASL Tutor	*"Different signing speeds should still be accepted."*	Implemented **speed-tolerant gesture comparison** using motion trajectory matching.
Prototype Users	*"The system should work without complex hardware."*	Used **MediaPipe landmark extraction** for lightweight webcam-based processing.
Research Validation	*"Limited samples can cause overfitting."*	Removed dataset limits and retrained models using the **full merged dataset** for better generalization.



Panel Comment Received at Proposal

"Emphasise how the proposed methodology differs from a normal sign language tutoring system; what is the proposed sign language type?"

How CareSense Addresses This

- **Intelligent Evaluation:** Existing systems mainly provide correct or incorrect results with limited feedback. CareSense uses DTW and MediaPipe to support speed-tolerant gesture recognition and provide real-time corrective feedback for better learning.

- **Personalized Learning:** Traditional systems follow the same learning path for all users. CareSense adapts lessons based on user mistakes, tracks progress, and recommends personalized practice activities to improve learning outcomes.

- **Proposed Sign Language Type**: The proposed system focuses on American Sign Language (ASL). The selected ASL gestures are specially designed for caregiving, healthcare, and patient communication situations

Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery

COMMON SLIDES FOR THE RP FOLLOWS FROM HERE →

Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery

MARKET OPPORTUNITY Global healthcare-AI market projected to exceed **USD 180 billion by 2030**; caregiver-wellness sub-segment growing at **>30% CAGR** — Sri Lanka / South Asia: rapidly ageing population + rising elderly-care demand

1 Commercialization & Sustainability

- **SaaS subscription model** for hospitals & elder-care facilities; tiered pricing by bed count and modules enabled (stress, fall, ADL, sign-language)
- **Sustained through ongoing licensing, model-update services**, and partnerships with caregiver-training institutions for the sign-language module

2 Customer Persona

- **Primary buyers:** hospital administrators and nursing supervisors managing 50–500 caregiving staff; eldercare home operators; private home-care service providers
- **Pain points:** staff burnout-driven turnover, patient-fall liability, documentation overhead, and communication gaps with deaf / hard-of-hearing patients

TARGET MARKET Sri Lanka & South Asia — expanding private hospital sector + strong serviceable demand from elderly-care growth

3 What is Unique in Our Solution

- **Only integrated platform** addressing four critical pain points — caregiver stress, patient fall risk, automated documentation, and accessibility — in one ecosystem
- **Personalisation per caregiver:** Isolation Forest baselines, RL-based ASL tutoring, and explainability (SHAP) — not a one-size-fits-all black box
- **Edge-first, privacy-preserving design** — raw video never leaves the device; addresses a key adoption barrier in patient-facing AI

4 How Will the Cost Be Recovered

- **Recurring SaaS fees** (monthly / annual per facility) cover model updates and ongoing support
- **Modular per-component licensing** — facilities pay only for modules they enable, lowering the entry barrier
- **Additional revenue streams:** training programmes for the sign-language module, premium analytics dashboards, and research-grade anonymised data partnerships (with consent)

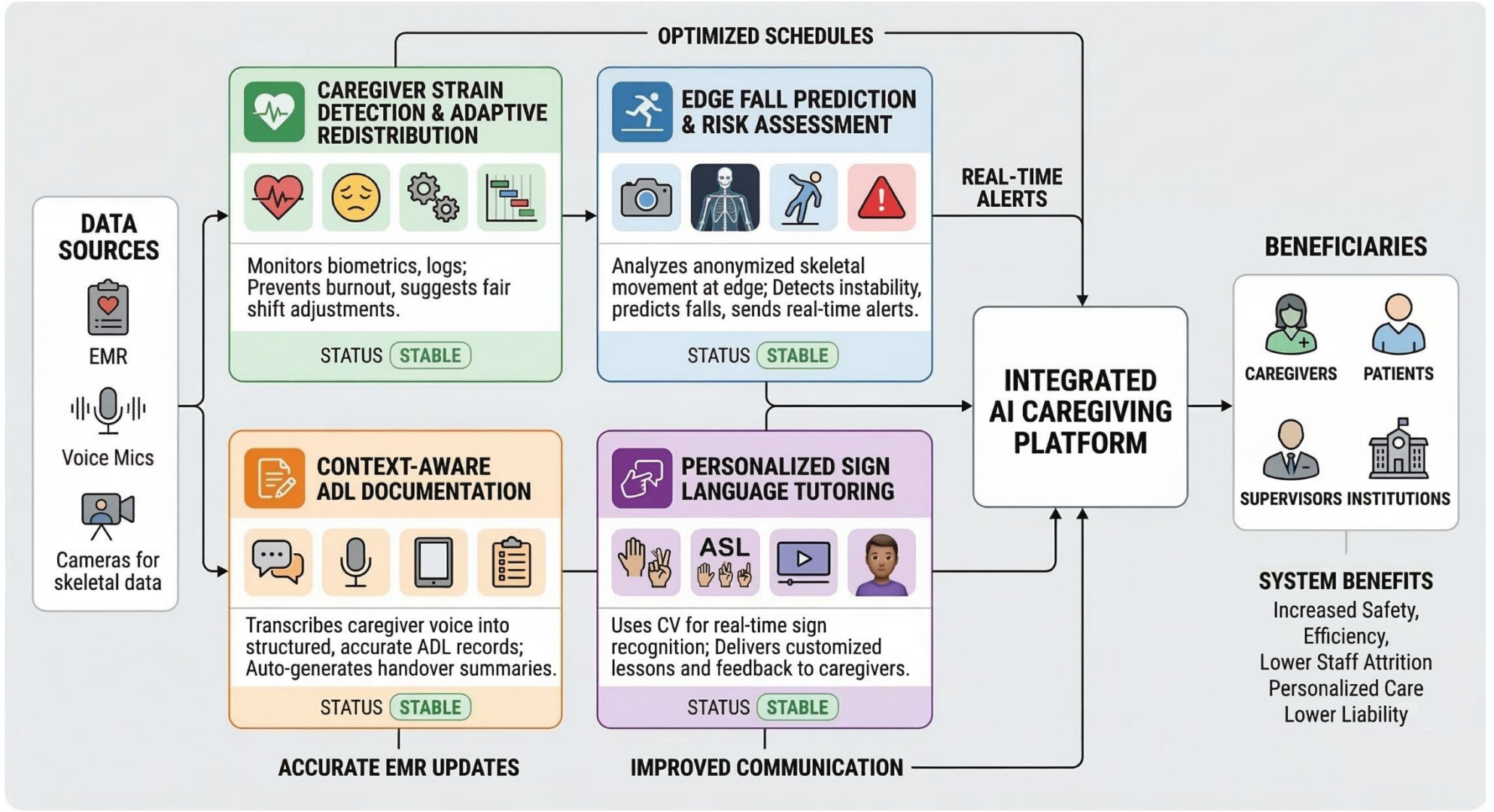
DEFENSIBILITY Four integrated AI modules + edge-first privacy stance create switching costs hard for single-feature competitors to match

Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery

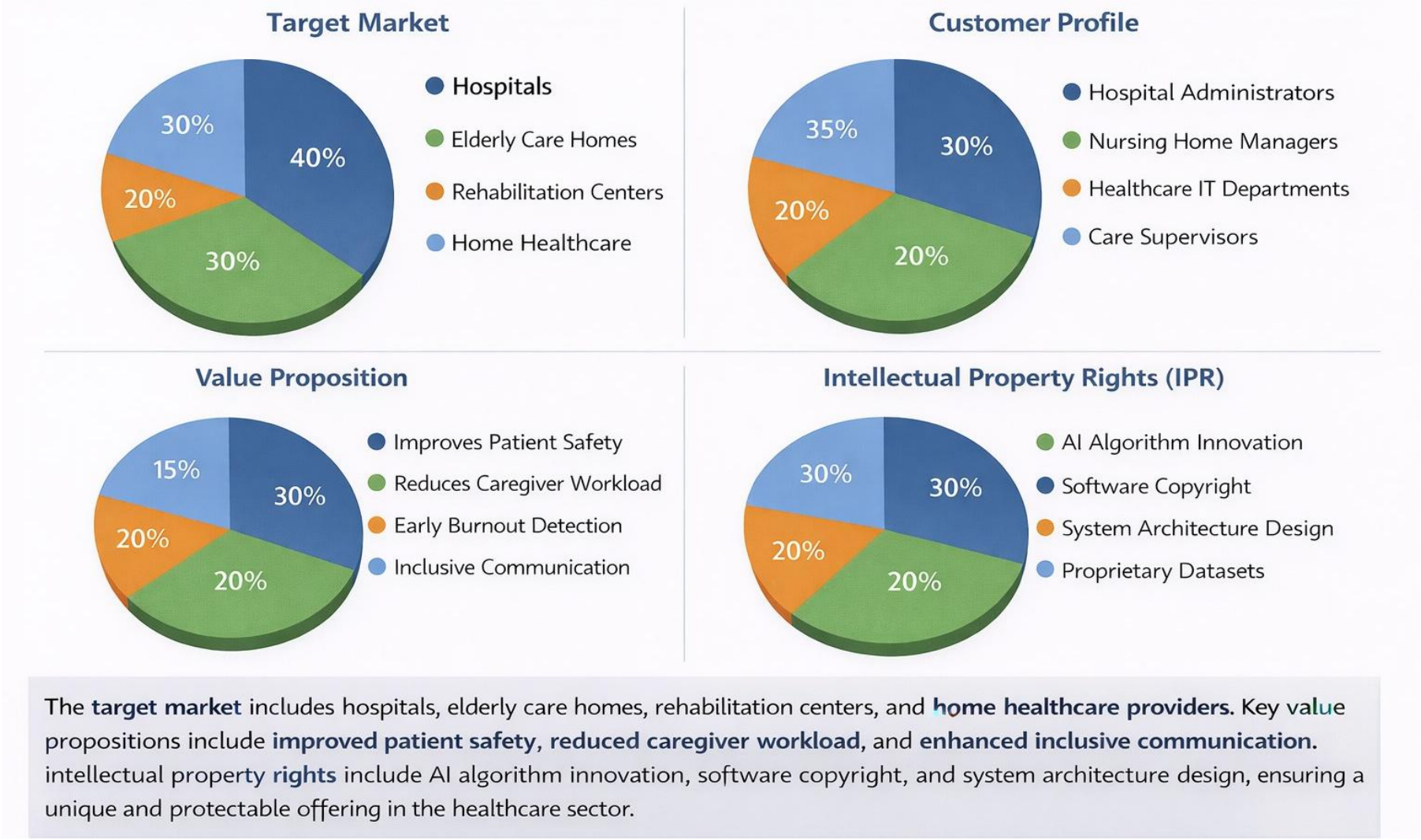
Ability of Commercialization – Key Points

Key Point	Description	
Integrated AI Healthcare Platform	The project can be commercialized as a unified AI-powered caregiver support platform combining monitoring, documentation, workforce analytics, and communication tools.	
High Demand in Healthcare Sector	Hospitals, elderly care homes, and rehabilitation centers face increasing demand for digital solutions to improve patient safety and caregiver efficiency.	
Improves Patient Safety	The real-time fall prediction system can significantly reduce patient injuries and liability risks, making it valuable for healthcare institutions.	
Reduces Caregiver Workload	Automated voice-based documentation and workload monitoring reduce administrative burden and improve caregiver productivity.	
Workforce Management Optimization	The caregiver deterioration detection module helps healthcare organizations optimize staff allocation and prevent burnout.	
Supports Incllisive Healthcare Communication	The sign language tutoring system enables caregivers to communicate effectively with deaf or hard-of-hearing patients.	**Privacy-Preserving Monitoring Technology** The skeletal pose-based fall detection ensures patient privac, increasing acceptance in healthcare environments with strict data protection requirements.
Scalable Deployment Model	The platform can be deployed across multiple healthcare facilities through cloud or edge-based infrastructure.	**Multiple Revenue Opportunities** The solution can be commercialized through SaaS subscriptions, enterprise licensing, or healthcare technology partnerships.

Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery – System Overview Diagram

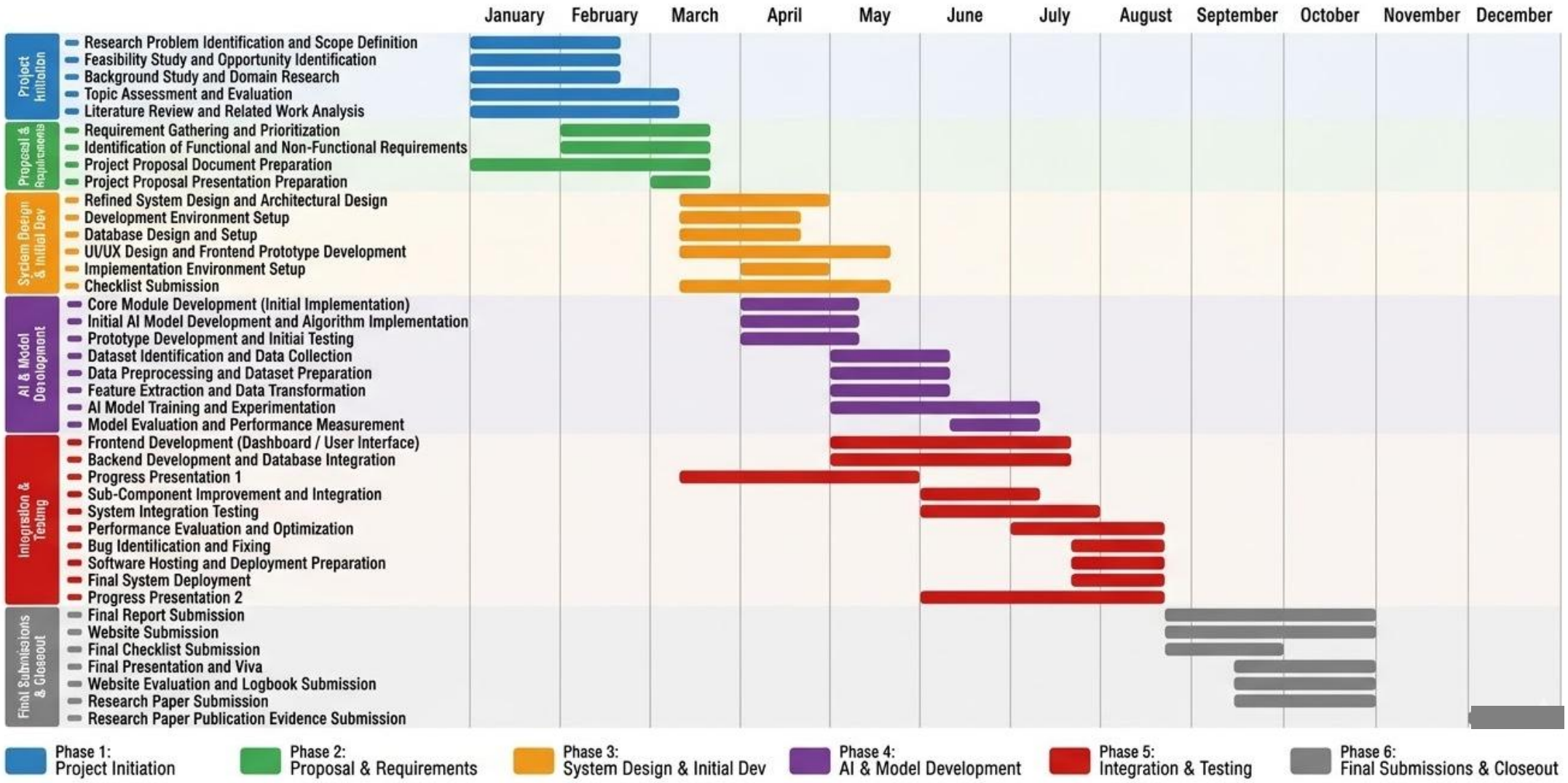


Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery



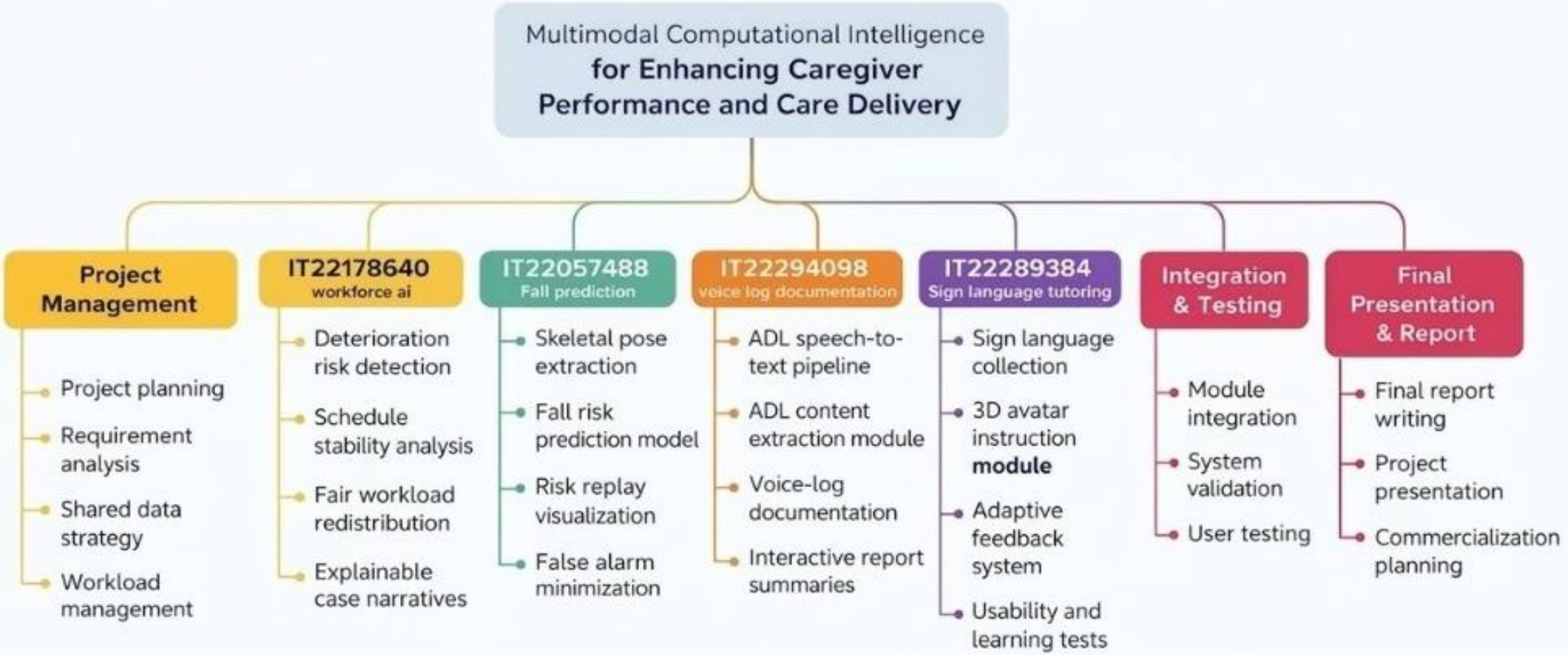
The **target market** includes hospitals, elderly care homes, rehabilitation centers, and **home healthcare providers.** Key value propositions include **improved patient safety, reduced caregiver workload,** and **enhanced inclusive communication.** intellectual property **rights** include AI algorithm innovation, software copyright, and system architecture design, ensuring a unique and protectable offering in the healthcare sector.

Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery – Gantt Chart



Work Breakdown Structure

*Multimodal Computational Intelligence for Enhancing **Caregiver Performance** and **Care Delivery***




Q&A


Thank You!